Universidad Carlos III de Madrid

Bachelor in Aerospace Engineering

Mechanics Applied to Aerospace Engineering

Particle on Oscillating Loop

Laboratory Session 2

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1 Introduction

In this laboratory report, we will investigate the dynamics of a heavy point particle P of mass m that slides without friction on a circular wire of centre C and radius R. The wire is on a vertical plane Oxy that rotates about the vertical axis Oy with a known angular velocity $\mathbf{\Omega}(t) = A\omega \sin(\omega t)\, \mathbf{j}_1$, and the position of P on the wire is given by the angle φ between $\overrightarrow{CP}$ and Ox (Fig. 1).

The main difficulty is that the frame where the motion can be easily described, the one attached to the wire, is not inertial and does not rotate at a constant rate, so the centrifugal, Euler and Coriolis forces all act on the particle. Our objective is to derive the equation of motion and the reaction of the wire, to integrate it in Matlab with `ode45` and explain the forces and the energy we obtain, to find a condition that tells if the particle oscillates or rotates on a fixed loop and check it with simulations, and to see how far we can trust a linearized equation around the lowest point of the wire, where resonance appears.

In Section 2 we derive the analytical model and in Section 3 we explain how we have implemented it in Matlab. In Section 4 we present the results and discuss each of them next to its figure, and Section 5 gives the conclusions.

2 Analytical Model

2.1 Reference Frames and Degrees of Freedom

$S_1 = Oxyz$ rotates with the plane of the wire: $\mathbf{i}_1$ is horizontal along the diameter through C, $\mathbf{j}_1$ is vertical and upward (the rotation axis) and $\mathbf{k}_1 = \mathbf{i}_1 \times \mathbf{j}_1$ is normal to the plane, pointing out of the paper in Fig. 1. $S_0 = Ox_0y_0z_0$ is a stationary frame, assumed inertial, that coincides with S_1 at $t = 0$, with $\mathbf{j}_0 = \mathbf{j}_1$. The axis Oy is tangent to the wire at O, so $\overrightarrow{OC} = R\,\mathbf{i}_1$.

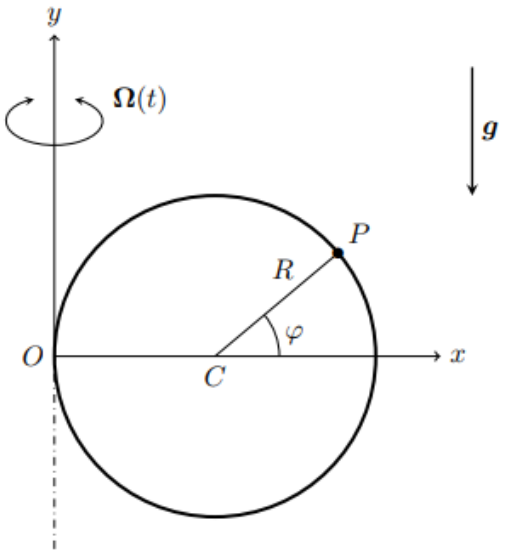


Figure 1: Geometry of the problem.

In space, P needs three coordinates. The wire removes two of them, since P has to stay in the plane Oxy and at a distance R from C, which means that the problem only has $3 - 2 = 1$ configuration degree of freedom (φ). The position of P seen from the stationary frame depends on φ and also explicitly on t.

The angle θ turned by the plane satisfies $\dot\theta = \Omega$ with $\theta(0) = 0$, where $\Omega = A\omega \sin \omega t$. Integrating with the change $u = \omega\tau$, $du = \omega\, d\tau$:

$$\theta(t) = \int_0^t A\omega \sin(\omega\tau)\, d\tau = A \int_0^{\omega t} \sin u\, du = A\big[-\cos u\big]_0^{\omega t} = A\,(1 - \cos \omega t), \tag{2.1}$$

so θ oscillates between $\theta = 0$ and $\theta = 2A$ with period $T = 2\pi/\omega$. Using this angle θ to get the rotating reference frame S_1, we obtain these unit vectors as functions of the stationary frame's unit vectors:

$$\mathbf{i}_1 = \cos\theta\, \mathbf{i}_0 - \sin\theta\, \mathbf{k}_0, \quad \mathbf{j}_1 = \mathbf{j}_0, \quad \mathbf{k}_1 = \sin\theta\, \mathbf{i}_0 + \cos\theta\, \mathbf{k}_0, \tag{2.2}$$

2.2 Position, Velocity and Acceleration of P

The side adjacent to φ is $R\cos\varphi$ along $\mathbf{i}_1$ and the opposite one is $R\sin\varphi$ along $\mathbf{j}_1$. Then,

$$\mathbf{r}_1^P = \overrightarrow{OC} + \overrightarrow{CP} = R(1+\cos\varphi)\,\mathbf{i}_1 + R\sin\varphi\,\mathbf{j}_1. \tag{2.3}$$

Using the auxiliary polar basis defined as:

$$\mathbf{e}_r = \cos\varphi\,\mathbf{i}_1 + \sin\varphi\,\mathbf{j}_1, \quad \mathbf{e}_\varphi = -\sin\varphi\,\mathbf{i}_1 + \cos\varphi\,\mathbf{j}_1, \tag{2.4}$$

With the time derivatives of the unit vectors being:

$$\dot{\mathbf{e}}_r = \dot\varphi\,\mathbf{e}_\varphi, \quad \dot{\mathbf{e}}_\varphi = -\dot\varphi\,\mathbf{e}_r, \tag{2.5}$$

Also,

$$\mathbf{i}_1 = \cos\varphi\,\mathbf{e}_r - \sin\varphi\,\mathbf{e}_\varphi, \quad \mathbf{j}_1 = \sin\varphi\,\mathbf{e}_r + \cos\varphi\,\mathbf{e}_\varphi, \tag{2.6}$$

Differentiating in the S_1 reference frame applying the chain rule, we get the velocity and acceleration:

$$\mathbf{v}_1^P = -R\dot\varphi\sin\varphi\,\mathbf{i}_1 + R\dot\varphi\cos\varphi\,\mathbf{j}_1 = R\dot\varphi\,\mathbf{e}_\varphi, \tag{2.7}$$

$$\mathbf{a}_1^P = -R(\ddot\varphi\sin\varphi + \dot\varphi^2\cos\varphi)\,\mathbf{i}_1 + R(\ddot\varphi\cos\varphi - \dot\varphi^2\sin\varphi)\,\mathbf{j}_1 = R\ddot\varphi\,\mathbf{e}_\varphi - R\dot\varphi^2\,\mathbf{e}_r. \tag{2.8}$$

2.3 Forces on the Particle

Two real forces act on P: its weight and the reaction of the wire. The wire is frictionless, so the reaction has no component along the tangent $\mathbf{e}_\varphi$:

$$m\mathbf{g} = -mg\,\mathbf{j}_1 = -mg\,(\sin\varphi\,\mathbf{e}_r + \cos\varphi\,\mathbf{e}_\varphi), \qquad \mathbf{N} = N_1\,\mathbf{e}_r + N_2\,\mathbf{k}_1. \tag{2.9}$$

Since S_1 is not inertial, we have to add the inertia forces. With $\mathbf{j}_1\times\mathbf{i}_1 = -\mathbf{k}_1$, $\mathbf{j}_1\times\mathbf{j}_1 = \mathbf{0}$ and $\mathbf{j}_1\times\mathbf{k}_1 = \mathbf{i}_1$ we get $\boldsymbol{\Omega}\times\mathbf{r}_1^P = -\Omega R(1+\cos\varphi)\,\mathbf{k}_1$, $\dot{\boldsymbol{\Omega}}\times\mathbf{r}_1^P = -\dot\Omega R(1+\cos\varphi)\,\mathbf{k}_1$, $\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\mathbf{r}_1^P) = -\Omega^2 R(1+\cos\varphi)\,\mathbf{i}_1$ and $\boldsymbol{\Omega}\times\mathbf{v}_1^P = \Omega R\dot\varphi\sin\varphi\,\mathbf{k}_1$. We also use $\mathbf{i}_1 = \cos\varphi\,\mathbf{e}_r - \sin\varphi\,\mathbf{e}_\varphi$:

$$\begin{aligned}\mathbf{F}_{cent} &= -m\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\mathbf{r}_1^P) = m\Omega^2 R(1+\cos\varphi)\,\mathbf{i}_1 \\ &= mA^2\omega^2 R\sin^2(\omega t)(1+\cos\varphi)\,[\cos\varphi\,\mathbf{e}_r - \sin\varphi\,\mathbf{e}_\varphi],\end{aligned} \tag{2.10}$$

$$\mathbf{F}_{Euler} = -m\dot{\boldsymbol{\Omega}}\times\mathbf{r}_1^P = m\dot\Omega R(1+\cos\varphi)\,\mathbf{k}_1 = mA\omega^2 R\cos(\omega t)(1+\cos\varphi)\,\mathbf{k}_1, \tag{2.11}$$

$$\mathbf{F}_{Cor} = -2m\boldsymbol{\Omega}\times\mathbf{v}_1^P = -2mA\omega R\dot\varphi\sin(\omega t)\sin\varphi\,\mathbf{k}_1. \tag{2.12}$$

2.4 Equation of Motion and Reaction Forces

In S_1, $m\mathbf{a}_1^P = m\mathbf{g} + \mathbf{N} + \mathbf{F}_{cent} + \mathbf{F}_{Euler} + \mathbf{F}_{Cor}$. The direction $\mathbf{e}_\varphi$ contains no reaction, and the Euler and Coriolis forces are along $\mathbf{k}_1$, so projecting on $\mathbf{e}_\varphi$ gives $mR\ddot\varphi = -mg\cos\varphi - m\Omega^2 R(1+\cos\varphi)\sin\varphi$, that is:

$$\boxed{\ddot\varphi = -\frac{g}{R}\cos\varphi - A^2\omega^2\sin^2(\omega t)\,(1+\cos\varphi)\sin\varphi} \tag{2.13}$$

The other two projections give the reactions. On $\mathbf{e}_r$: $-mR\dot\varphi^2 = N_1 - mg\sin\varphi + m\Omega^2 R(1+\cos\varphi)\cos\varphi$, and on $\mathbf{k}_1$: $0 = N_2 + m\dot\Omega R(1+\cos\varphi) - 2m\Omega R\dot\varphi\sin\varphi$:

$$N_1 = -mR\dot\varphi^2 + mg\sin\varphi - mA^2\omega^2 R\sin^2(\omega t)(1+\cos\varphi)\cos\varphi, \tag{2.14}$$

$$N_2 = mA\omega R\big[2\dot\varphi\sin(\omega t)\sin\varphi - \omega\cos(\omega t)(1+\cos\varphi)\big]. \tag{2.15}$$

2.5 Mechanical Energy and Work of the Wire

We evaluate the mechanical energy in the stationary frame S_0 rather than in S_1. Because $\boldsymbol{\Omega}(t)$ varies with time ($\dot{\boldsymbol{\Omega}} \neq \mathbf{0}$), the centrifugal force has no stationary potential and the Euler force is non-conservative. In S_0, only real forces do work.

The absolute velocity is $\mathbf{v}_0^P = \dot{\mathbf{r}}_1^P + \boldsymbol{\Omega}\times\mathbf{r}_1^P = \mathbf{v}_1^P + \boldsymbol{\Omega}\times\mathbf{r}_1^P = R\dot\varphi\,\mathbf{e}_\varphi - \Omega R(1+\cos\varphi)\,\mathbf{k}_1$. Since $\mathbf{e}_\varphi \perp \mathbf{k}_1$, $|\mathbf{v}_0^P|^2 = R^2\dot\varphi^2 + \Omega^2 R^2(1+\cos\varphi)^2$, so the kinetic energy is $K_0^P = \frac{1}{2}m|\mathbf{v}_0^P|^2$, and taking the height of O as the reference for the potential energy,

$$K_0^P = \tfrac{1}{2}m\big(R^2\dot\varphi^2 + \Omega^2 R^2(1+\cos\varphi)^2\big), \qquad V_0^P = mgR\sin\varphi, \qquad E_0^P = K_0^P + V_0^P. \tag{2.16}$$

By the work–energy theorem $dK_0^P/dt = (m\mathbf{g} + \mathbf{N})\cdot\mathbf{v}_0^P$, because in the stationary frame they are the only forces acting on P, and the weight is conservative, $m\mathbf{g}\cdot\mathbf{v}_0^P = -dV_0^P/dt$. Only the power of the

reaction is left, and since $\mathbf{e}_r \perp \mathbf{e}_\varphi$ and $\mathbf{e}_r \perp \mathbf{k}_1$,

$$\frac{dE_0^P}{dt} = \mathbf{N}\cdot\mathbf{v}_0^P = (N_1\mathbf{e}_r + N_2\mathbf{k}_1)\cdot(R\dot\varphi\,\mathbf{e}_\varphi - \Omega R(1+\cos\varphi)\,\mathbf{k}_1) = -\Omega R(1+\cos\varphi)N_2. \tag{2.17}$$

On a fixed wire the reaction is perpendicular to the velocity. Here the wire moves with the plane, so N_2 does work and E_0^P is conserved if and only if $A = 0$, as we can observe on Eq. (2.15).

2.6 Fixed Loop: Criterion for the Motion

When the wire is fixed ($A = 0$), the plane does not rotate ($\Omega = 0$), so the reaction N_2 does no work and the mechanical energy from Eq. (2.16) is strictly conserved ($dE_0^P/dt = 0$):

$$\frac{1}{2}\dot\varphi^2 + \frac{g}{R}\sin\varphi = \frac{1}{2}\dot\varphi_0^2 + \frac{g}{R}\sin\varphi_0 = \frac{E_0^P}{mR^2}. \tag{2.18}$$

The potential energy has its maximum, mgR, at the top of the loop ($\varphi = \pi/2$). If $E_0^P > mgR$, P reaches the top with $\dot\varphi \neq 0$, so $\dot\varphi$ never changes sign and φ grows (or decreases) steadily. If $E_0^P < mgR$, P stops before the top and oscillates. Imposing $E_0^P > mgR$ in Eq. (2.18), P rotates if

$$|\dot\varphi_0| > \sqrt{\frac{2g}{R}(1-\sin\varphi_0)}, \tag{2.19}$$

with $\dot\varphi_0 > 0$ for a growing φ. Otherwise, $\dot\varphi = 0$ in Eq. (2.18) gives the turning points, $\varphi_{\max} = \arcsin(\sin\varphi_0 + R\dot\varphi_0^2/(2g))$ and $\varphi_{\min} = -\pi - \varphi_{\max}$. The limit case $E_0^P = mgR$ is the curve $\dot\varphi = \pm\sqrt{(2g/R)(1-\sin\varphi)}$ that separates the oscillations from the rotations, shown in Fig. 6.

2.7 Linearized Equation

With $\varphi = \delta - \pi/2$, $\cos\varphi = \sin\delta$, $\sin\varphi = -\cos\delta$ and $\ddot\varphi = \ddot\delta$, the Eq. (2.13) becomes the exact $\ddot\delta = -(g/R)\sin\delta + \Omega^2(1+\sin\delta)\cos\delta$. For $\delta \ll 1$, $\sin\delta \approx \delta$ and $\cos\delta \approx 1$, which gives $\ddot\delta = -(g/R)\delta + \Omega^2(1+\delta)$. The hypothesis $A^2 \ll g/(R\omega^2)$ means $\Omega^2 \leq A^2\omega^2 \ll g/R$, so $\Omega^2\delta$ is dropped against $(g/R)\delta$. Using $\sin^2\omega t = \frac{1}{2}(1-\cos 2\omega t)$:

$$\ddot\delta + \frac{g}{R}\delta - \Omega^2 = 0 \quad\Longleftrightarrow\quad \ddot\delta + \omega_n^2\delta = F_0\,(1-\cos 2\omega t), \qquad \omega_n = \sqrt{g/R}, \quad F_0 = \tfrac{1}{2}A^2\omega^2. \tag{2.20}$$

This is an undamped harmonic oscillator of natural frequency ω_n that is forced at the frequency 2ω by the centrifugal force, which reaches its maximum twice per period of the plane. We assume P at rest at the bottom, $\delta(0) = \dot\delta(0) = 0$. When the forcing frequency is equal to the natural one, $2\omega = \omega_n$, that is $\omega = \frac{1}{2}\sqrt{g/R}$, we have resonance: the centrifugal force pushes the particle at the same rhythm as its natural oscillation. This makes the oscillation get larger and larger, so δ grows with time and the hypothesis $\delta \ll 1$ stops being true.

3 Numerical Implementation in Matlab

Equation (2.13) has to be written for the state vector $X = [\varphi, \dot\varphi]^T = [X_1, X_2]^T$ because `ode45` only integrates first-order systems:

$$\frac{d}{dt}\begin{pmatrix}X_1\\X_2\end{pmatrix} = f(t,X) = \begin{pmatrix} X_2 \\ -\dfrac{g}{R}\cos X_1 - A^2\omega^2\sin^2(\omega t)\,(1+\cos X_1)\sin X_1\end{pmatrix}. \tag{3.1}$$

This is the function `diffeq(t,X,g,omega,R,A)` of `main.m`, and Eq. (2.20) is `diffeqlin`, with $X = [\delta, \dot\delta]^T$. The options are `RelTol` $= 10^{-8}$, `AbsTol` $= 10^{-10}$ (the default 10^{-6} is not small against $\delta \sim 10^{-3}$) and `MaxStep` $= 0.01$ s, so every swing has sufficient points. We compute the forces and energies afterwards from φ and $\dot\varphi$ with Eqs. (2.10)–(2.15) and (2.16). The position in the stationary frame follows from Eqs. (2.1)–(2.3): $x_0 = R(1+\cos\varphi)\cos\theta$, $y_0 = R\sin\varphi$, $z_0 = -R(1+\cos\varphi)\sin\theta$. A fixed-loop case counts as rotating if $\dot\varphi$ never changes sign and as oscillating otherwise, and the periods come from the times of the extrema of φ.

In all cases $g = 9.81\,\mathrm{m/s^2}$ and $m = 1\,\mathrm{kg}$. The oscillating loop uses $A = \pi$, $R = 1\,\mathrm{m}$, $\omega = 0.1\,\mathrm{rad/s}$, $\varphi(0) = \dot\varphi(0) = 0$ and $t \in [0, 3T]$ with $T = 62.83\,\mathrm{s}$. The fixed loop uses $A = 0$ and $R = 1\,\mathrm{m}$, and the linearized study $A = 0.1$, $R = 0.01\,\mathrm{m}$ and $\omega_n = 31.32\,\mathrm{rad/s}$. The guide do not fix the duration of these two studies, so we take $t \in [0, 10]$ s, which is longer than any period of the fixed loop.

4 Results and Discussion

4.1 Angle and Reaction Forces

Figure 2 shows φ, N_1 and N_2 for $t \in [0, 3T]$, with $T = 62.83\,\text{s}$. There are too many swings in that interval to tell them apart, so the right panels zoom into a window of 10 s around $t = T/4$. At $t = 0$ the particle is at rest at the end of the horizontal diameter, so it falls and swings around the bottom of the wire, between $\varphi \approx 0$ and $\varphi \approx -\pi$ ($\varphi \in [-3.14, 0.01]$ rad), passing through O at the left end. The motion is almost the one of a simple pendulum, and its amplitude does not change during the three periods of the plane, because $\omega = 0.1\,\text{rad/s}$ is small: the plane turns slowly and the centrifugal term of Eq. (2.13) is small compared with g/R. For the same reason the period of the swing, 2.37 s, is the one of a pendulum with a large amplitude (90° to each side of the bottom), and it is longer than the period of small oscillations, $2\pi\sqrt{R/g} = 2.01\,\text{s}$, because for a large amplitude the restoring force grows less than proportionally to the angle from the bottom, so the pendulum is slower.

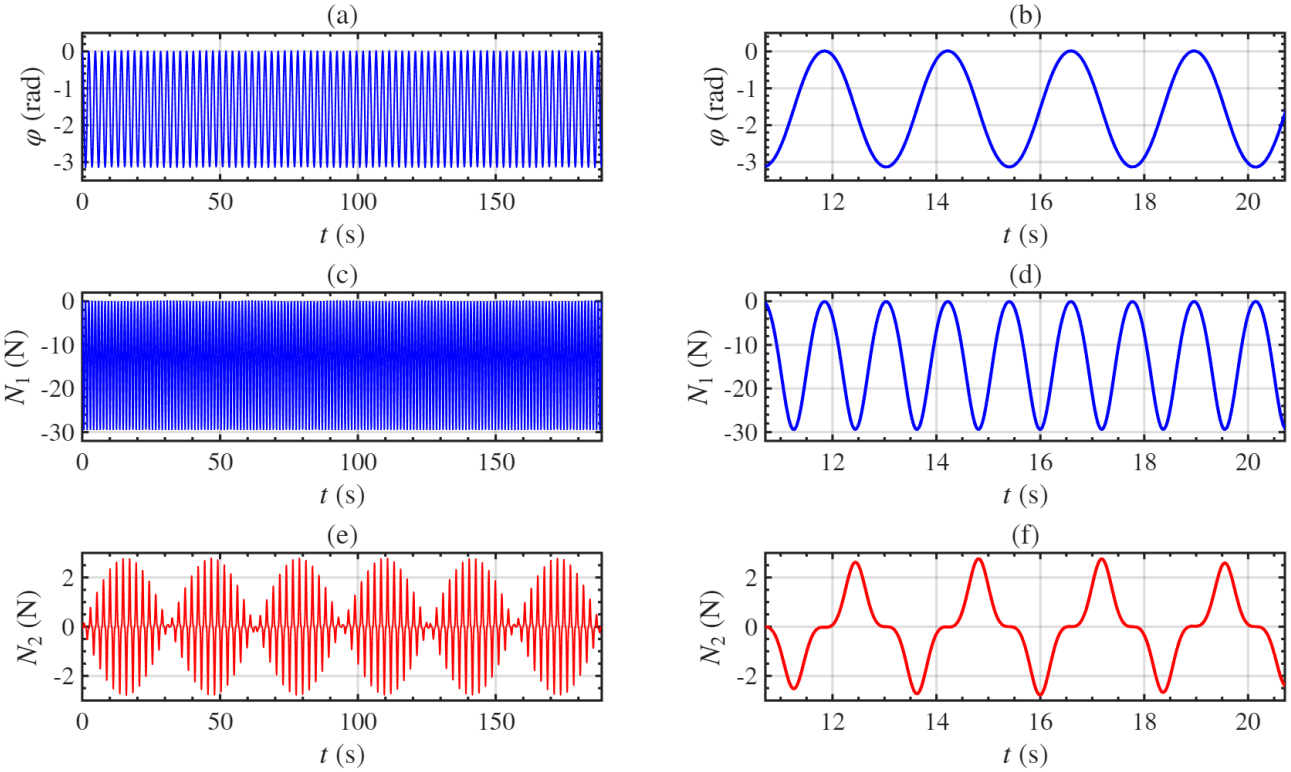


Figure 2: Oscillating loop. Left: $t \in [0, 3T]$. Right: window around $t = T/4$. (a, b) Angle; (c, d) reaction along $\mathbf{e}_r$; (e, f) reaction along $\mathbf{k}_1$.

N_1 is always negative, between $-29.43\,\text{N}$ and $0.00\,\text{N}$, so the wire pulls P towards the centre C. It is zero at the turning points, where P stops, and it has its minimum at the bottom, where the speed is largest ($\dot{\varphi} = 4.43\,\text{rad/s}$). As the rotation is weak, Eq. (2.18) is approximately valid and gives $\dot{\varphi}^2 \approx 2g/R$ at the bottom, so Eq. (2.14) gives $N_1 \approx -2mg - mg = -3mg$: the wire balances the weight and also pulls P towards C to make it turn. The particle crosses the bottom twice per swing, so N_1 oscillates at twice the frequency of φ (Fig. 2d). N_2 is smaller, with a maximum of 2.78 N, and its sign changes from one crossing of the bottom to the next because P goes through it in opposite directions (Fig. 2f). Its amplitude changes with a period $T/2$ and it is small when the plane stops and turns back ($\Omega \approx 0$). We explain where N_2 comes from in the next section.

4.2 Inertia Forces

Figure 3 shows the three inertia forces. Each has its own axis because their sizes are different, as the guidelines recommend. The centrifugal force (a) always points away from the rotation axis and its maximum is 0.20 N, reached when the plane turns fastest and P is far from the axis. The Euler force (b) is proportional to $\dot{\Omega}$, so it follows $\cos \omega t$, and its maximum (0.06 N) is when the plane stops and changes its direction, which is when the centrifugal and Coriolis forces vanish. The Coriolis force (c)

is the largest, up to 2.78 N, because it depends on the velocity of P along the wire, $\dot{\varphi}$, which is large while the plane turns slowly. It vanishes when the plane stops and at the turning points, where $\dot{\varphi} = 0$.

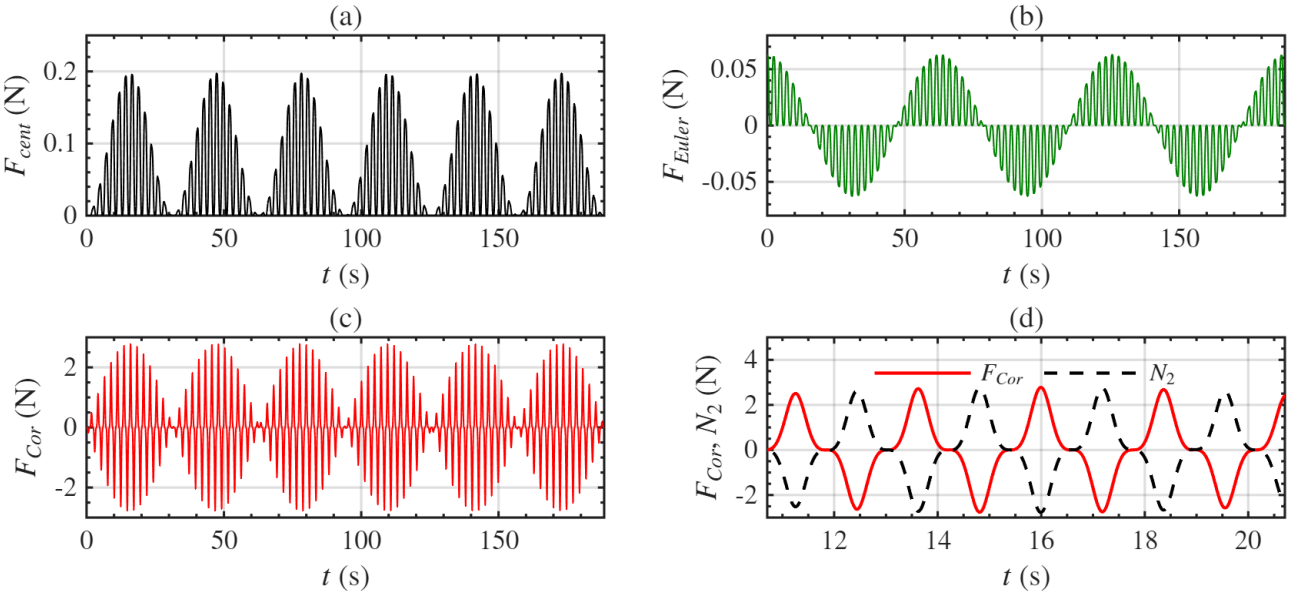


Figure 3: Inertia forces of the oscillating loop. (a) Centrifugal (along $\mathbf{i}_1$), (b) Euler and (c) Coriolis (along $\mathbf{k}_1$) for $t \in [0, 3T]$. (d) Coriolis force and reaction N_2 around $t = T/4$.

Only the centrifugal force has a component along $\mathbf{e}_\varphi$ and enters the equation of motion, Eq. (2.13), which is why φ moves almost as in a pendulum. The Euler and Coriolis forces are along $\mathbf{k}_1$, normal to the plane of the wire, so they do not move P along the wire. They are balanced by N_2 (Eq. (2.15)), and this is why N_2 in Fig. 3d is almost the opposite of F_{Cor}, the Euler force being small.

4.3 Energy

In Fig. 4(a, b), K_0^P and V_0^P are exchanged in each swing: K_0^P goes from almost zero at the turning points to its maximum, 9.83 J, at the bottom, where V_0^P is minimum (−9.81 J), as in a pendulum. The total energy is not constant (Fig. 4c, d): $E_0^P - E_0^P(0)$ varies between −0.08 J and 0.31 J. The reason is Eq. (2.17). The wire moves with the plane, so N_2 does work: when P moves away from the rotation axis the wire gives energy to P, and when P goes back to the axis the wire takes it back, so E_0^P is high when P is far from the axis and low when it is near (Fig. 4d). More precisely, $E_0^P - E_0^P(0)$ follows $m\Omega^2 R^2(1 + \cos\varphi)^2$ minus an offset of at most 0.08 J, which is the energy that the wire has spent speeding up the plane, and this is why E_0^P is below its initial value when P is near the axis. When the plane stops, $\Omega = 0$ and E_0^P is again almost its initial value: $E_0^P - E_0^P(0)$ is 0.001 J, 0.000 J and 0.001 J at T, $2T$ and $3T$. This is why the amplitude of φ does not change in Fig. 2a.

4.4 Trajectory in the Stationary Frame

Figure 5 shows the trajectory of P seen from the stationary frame. P always stays on the surface swept by the wire when it turns about the vertical axis, which is a torus without a hole because the wire touches the axis at O. Each swing draws one petal, from the axis ($\varphi = -\pi$) to the outer edge $2R$ ($\varphi = 0$) and back. The petals are spread around the axis because the plane turns from $\theta = 0$ to $\theta = 2\pi$ at $t = T/2$ and back to 0 at $t = T$ (Eq. (2.1)). Near $\theta = 0$ and $\theta = 2\pi$ the plane stops and reverses its rotation, so many petals accumulate in the same direction ($+x_0$ in Fig. 5b), while near $\theta = \pi$ the plane turns fastest (Ω is maximum) and the petals are more separated. The three periods of the plane draw almost the same figure, with slightly displaced petals because T is not a multiple of the swing period. The video `trajectory_video.avi`, generated by `main.m`, shows the first period.

4.5 Fixed Loop: Phase Portrait and Validation of the Criterion

Figure 6 shows $\dot{\varphi}$ as a function of φ for the nine initial conditions, together with the limit curve of Eq. (2.19), which separates the two kinds of motion. Since φ is plotted modulo 2π, both ends of the horizontal axis are the top of the loop. Table 1 compares the criterion of Eq. (2.19) with the simulation

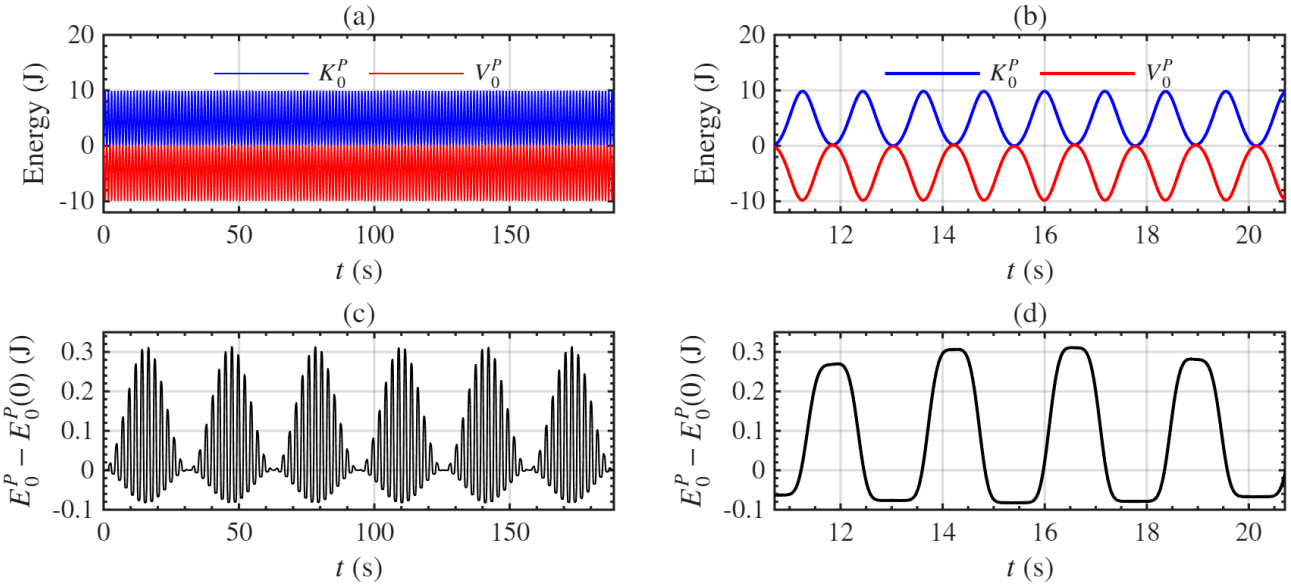


Figure 4: Energy of the oscillating loop. (a, b) Kinetic and gravitational potential energy; (c, d) change of the mechanical energy. Left: $t \in [0, 3T]$. Right: window around $t = T/4$.

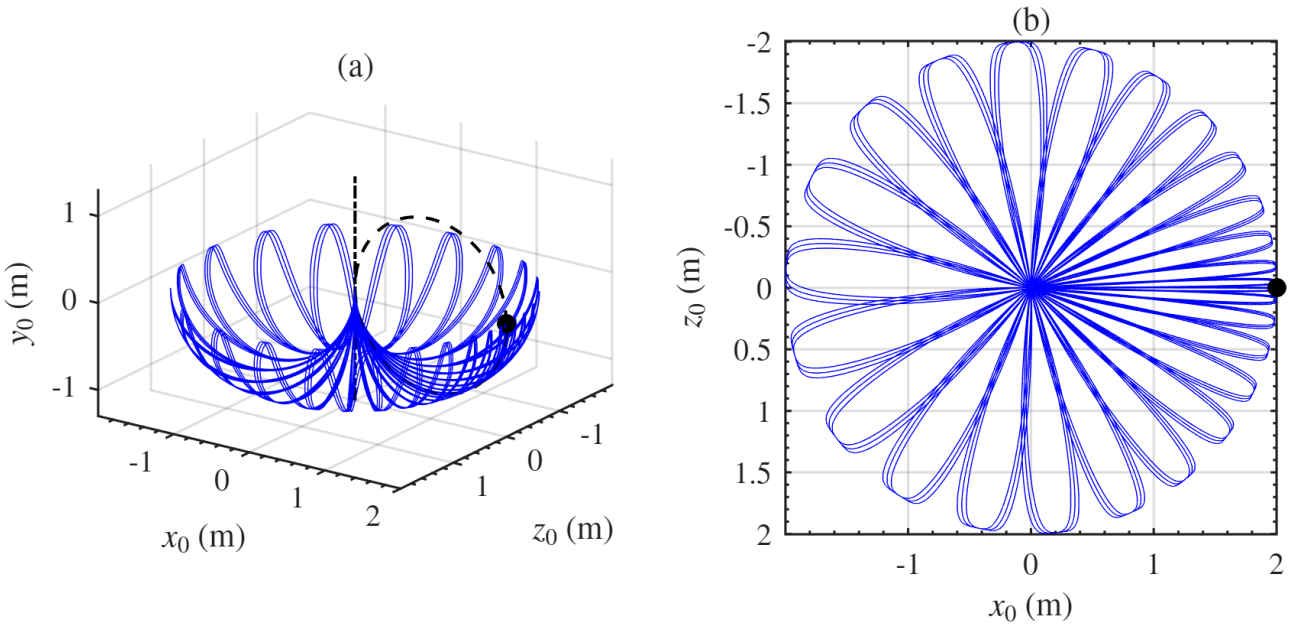


Figure 5: Trajectory of P in S_0 for $t \in [0, 3T]$, with the start marked by a dot. (a) Three-dimensional view with the wire at $t = 0$ (dashed) and the rotation axis (dash-dot). (b) View from above.

(we consider that P rotates if $\dot{\varphi}$ never changes sign), and they agree in all nine cases.

Table 1: Criterion of Eq. (2.19) against the simulation of the fixed loop.

φ_0 (rad)	$\dot{\varphi}_0$ (rad/s)	$\dot{\varphi}_c$ (rad/s)	Prediction	Simulation
0	0	4.429	oscillates	oscillates
0	1.5	4.429	oscillates	oscillates
0	3	4.429	oscillates	oscillates
$\pi/4$	0	2.397	oscillates	oscillates
$\pi/4$	1.5	2.397	oscillates	oscillates
$\pi/4$	3	2.397	rotates	rotates
$\pi/2$	0	0.000	rest	rest
$\pi/2$	1.5	0.000	rotates	rotates
$\pi/2$	3	0.000	rotates	rotates

The closed curves, around the stable equilibrium $(-\pi/2, 0)$, are oscillations. The open curves are the cases that start above the dash-dot curve, $(\pi/4, 3)$, $(\pi/2, 1.5)$ and $(\pi/2, 3)$. In these cases P has enough energy to pass the top of the loop ($E_0^P > mgR$), $\dot{\varphi}$ never becomes zero and φ increases steadily,

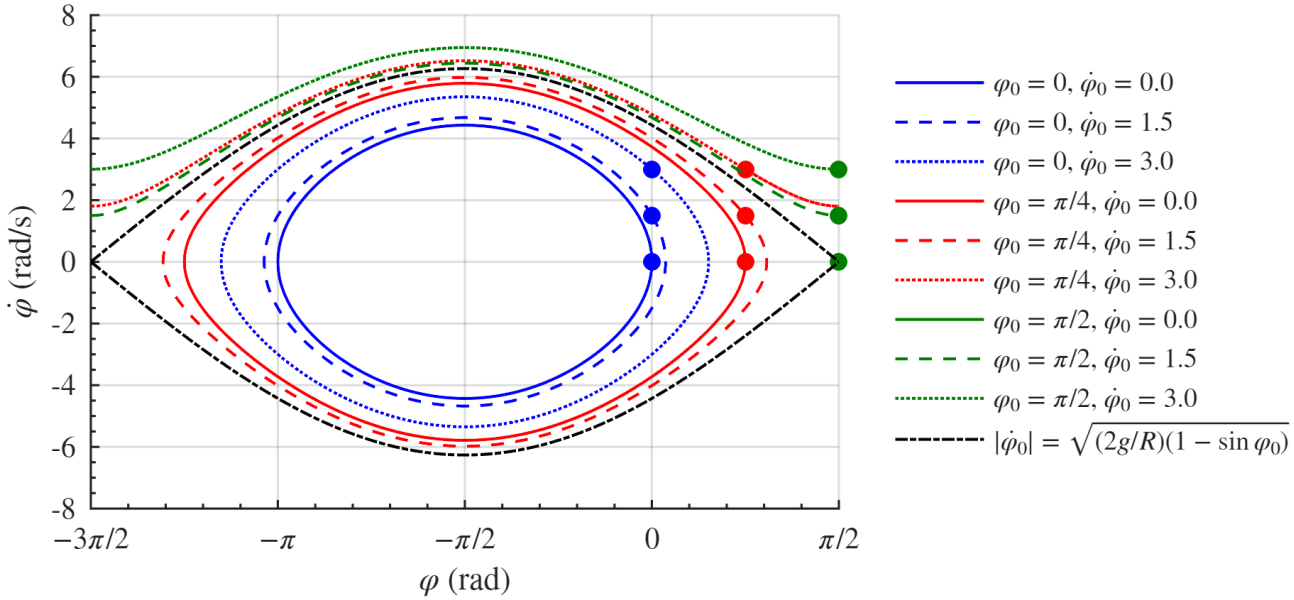


Figure 6: Phase portrait of the fixed loop ($A = 0$), with φ_0 in rad and $\dot{\varphi}_0$ in rad/s. Dots mark the initial conditions and the dash-dot line is the limit case $E_0^P = mgR$. Closed curves are oscillations and open curves are rotations.

with P going slowest at the top and fastest at the bottom. The speed alone is not enough: $(0, 3)$ has the same $\dot{\varphi}_0$ as $(\pi/4, 3)$ but less potential energy, and it oscillates. The period of the oscillations grows from 2.37 s to 3.36 s as the curves get closer to the dash-dot curve, because P spends more time near the top, where it moves slowly. The case $(\pi/2, 0)$ starts at rest at the top of the loop, which is an unstable equilibrium position ($\ddot{\varphi} = 0$), so it stays at rest in the simulation, although any disturbance would make it fall.

4.6 Linearized Against Full Equation

Figure 7a shows the percentage error between the linearized and the full solution for the three values of ω,

$$\text{Error} = 100\,\frac{|\delta_{lin} - \delta_{full}|}{\max|\delta_{full}|}.$$

We divide by the maximum of δ_{full} and not by $\delta(t)$ because δ crosses zero. The error oscillates fast, so the thick lines show its envelope. Figure 7b shows $\delta(t)$ for $\omega = \omega_n/2$.

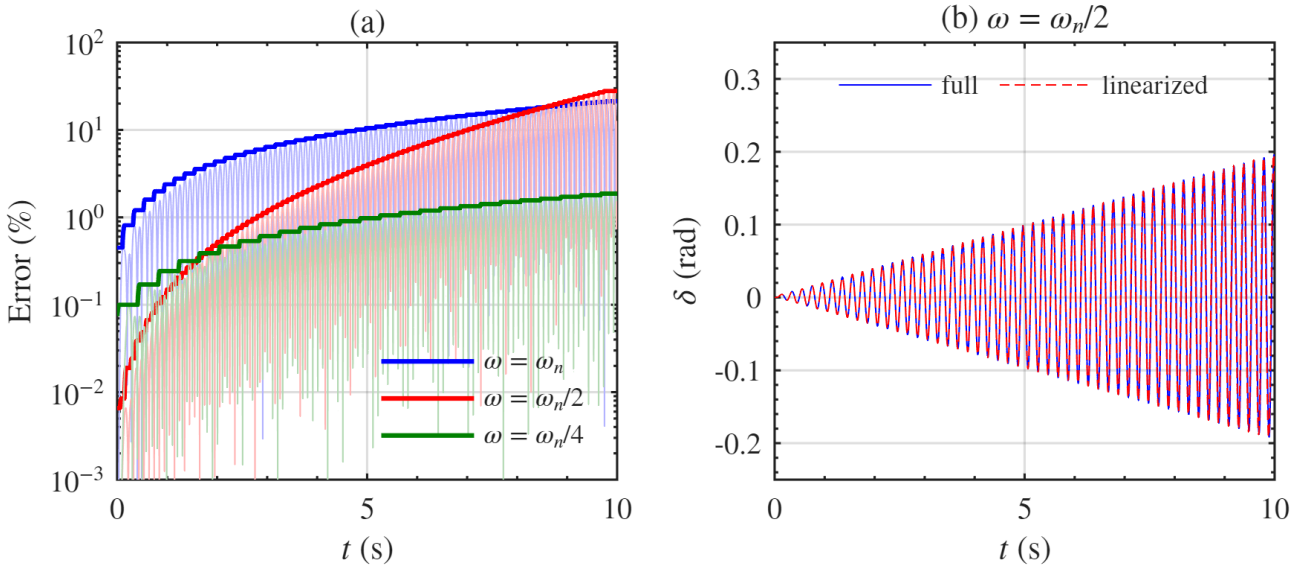


Figure 7: (a) Percentage error between the linearized and the full solutions (thin: error; thick: its envelope). (b) $\delta(t)$ of the full (solid) and linearized (dashed) solutions for $\omega = \omega_n/2$.

For $\omega = \omega_n/2$ there is resonance: the frequency of the centrifugal force, 2ω, is equal to the natural

frequency, $\omega_n = \sqrt{g/R}$, so each push is in phase with the oscillation and the amplitude of δ grows linearly with time (Fig. 7b), up to 0.20 rad ($\approx 11°$) at 10 s. Then the hypothesis $\delta \ll 1$ is not true any more, $\sin\delta \approx \delta$ fails and the error grows up to 27.8%. In the full equation the pendulum is slower when its amplitude is larger, so it gets out of phase with the forcing and the two solutions separate.

Outside resonance, the error is smaller for $\omega = \omega_n/4$ (below 2%) than for $\omega = \omega_n$ (21.4%), although δ stays small in both cases (at most about 0.013 rad). In the linearization we neglected $\Omega^2\delta$ against $(g/R)\delta$ using the hypothesis $A^2 \ll g/(R\omega^2)$, that is $A^2\omega^2 \ll g/R$. Since $\Omega^2 \propto \omega^2$, Ω^2 is 16 times larger at ω_n than at $\omega_n/4$, and δ is also larger, so the neglected term $\Omega^2\delta$ is even larger and it accumulates a larger phase difference between the two solutions as time passes.

5 Conclusions

- The system possesses a single configuration degree of freedom, φ. Projecting Newton's second law onto the moving polar frame yields a second-order differential equation for φ depending on gravity and centrifugal force.
- In the stationary frame, P traces a three-dimensional path confined to a torus swept by the loop. The trajectory looks like a flower with petals that condense around $+x_0$ near $\theta = 0$ and 2π, where the angular velocity vanishes ($\Omega \approx 0$), and spread out around $\theta = \pi$, where the plane rotates the fastest.
- The mechanical energy E_0^P is not conserved because the rotating wire does work on the particle at a rate $dE_0^P/dt = -\Omega R(1+\cos\varphi)N_2$. The force making the plane rotate with its angular velocity feeds energy into P as it comes closer or goes further from the axis. Because $\Omega(t)$ is periodic, E_0^P returns almost to its initial value every time the plane stops ($\Omega = 0$), for example at T, $2T$ and $3T$.
- On a stationary loop ($A = 0$), the $\dot\varphi$ and φ diagram depending on their initial values confirms the condition $|\dot\varphi_0| > \sqrt{(2g/R)(1-\sin\varphi_0)}$ we got analytically for the relation between $\dot\varphi$ and φ that determines whether the particle will oscillate or rotate around the wire. Rotation happens when $|\dot\varphi_0| > \sqrt{(2g/R)(1-\sin\varphi_0)}$. Oscillation happens when $|\dot\varphi_0| < \sqrt{(2g/R)(1-\sin\varphi_0)}$. When $|\dot\varphi_0| = \sqrt{(2g/R)(1-\sin\varphi_0)}$, the point reaches the top of the wire and stays there. As observed in our simulations, initial speed is not the only factor that determines the point's motion. For example, the particle that starts at $\varphi_0 = 0$ oscillates, while the one that starts at $\varphi_0 = \pi/4$ rotates, despite having the same initial velocity $\dot\varphi_0 = 3$ rad/s. This is due to the second one starting higher up (with more potential energy).
- The linearized equation for the motion of the particle around its equilibrium position is accurate only when the value of the angle displacement with respect to the equilibrium position is low ($\delta \ll 1$) and when the system is not at resonance ($\omega \neq \omega_n/2 = \frac{1}{2}\sqrt{g/R}$). At resonance, the angular displacement amplitude grows bigger and bigger, causing the condition $\delta \ll 1$ to not be satisfied and causing the linear and full non-linear solutions to diverge by up to 27.8% at $t = 10$ s.
- We have found particularly surprising that even though Coriolis force does not influence the motion of the particle at all, (which is determined by gravity and the centrifugal force), as we have seen in 4.2, Coriolis force is the inertia force with the biggest magnitude. The reason why this happens is that Coriolis force acts perpendicular to the plane where the wire is contained (where the particle is constrained to stay), only causing a reaction force in the wire of equal magnitude and opposite direction, as can be seen in Fig. 3d.

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